backtrack.

Stuck on one step.
Lost in whole lessons.

Find the step. Get a focused Khan repair. Return to your task.

University of the Philippines Manila

The search can become the study session

01 WHAT THEY NEED TO FINISH

Solve this equation

$$x^2 + 7x + 12 = 0$$

They search “quadratics”.

02 WHAT MAY BE BLOCKING IT

Find the factor pair

Multiply to 12. Add to 7.

BACKTRACK checks the gap before choosing the repair.

A short repair. A clear way back.

1. **Try today’s task**
2. **Find a useful earlier step**
3. **Learn + practise with Khan**
4. **Try a fresh current task**

Your answer shapes the route. The repair fits the missing step.

Two wrong answers. Different help.

Tried x = 3, 4 → Check signs → fresh goal

Tried x = 2, 6 → Check factors → Fresh goal

The mistake points to the starting check.

Khan material, focused on the missing step

Khan Academy

1

Watch the key part

1:34 focused segment
Match sum and product

→

2

Try the step here

Use the idea with different numbers

→

3

Return to the goal

Solve a fresh quadratic equation

Short enough to use now. More Khan practice when they need it.

Back to the skill that stopped them

FRESH CURRENT-LEVEL TASK

$$x^2 + 11x + 28 = 0$$

$$x = -4 \text{ or } x = -7$$

New numbers. The same skill they came to recover.

Try the demo. Watch the route change.

Compare two wrong answers.
See why they get different repairs.

Try BACKTRACK

backtrack-learning.vercel.app

Scan to start

A few sample routes to help you picture BACKTRACK. The full product is still in development.

Can they still do it next week?

1 Stuck before the route

2 Pass a fresh task after practice

3 Pass again 7–14 days later

Retained reentry

Count everyone who started unable to do the task.

A five-month school pilot

80–120 learners across 2–3 cohorts

NOV	DEC	JAN	FEB	MAR
Review maps + baseline	First school cycle	Compare + improve	New educator + repeat	Follow-up + handover

November 2026–March 2027

Pilot targets, fitted to each school's access and timetable.

Does the route add value?

Khan practice

Selected by the teacher

Khan + BACKTRACK

Practice tied to the recovery route

The same fresh assessments

After practice, then 7–14 days later. Similar time and support.

The route is the reward

What the learner does		What changes
Know the step	→	Skip its review
Repair a gap	→	Move closer to the goal
Come back later	→	Keep earlier progress

Progress changes what you need to do next.

One routine. Many learning goals.

Fractions	Algebra	Graphs

Every new destination uses a reusable kit

A reviewed map + fresh checks + matched Khan material

Another educator can run the same recovery routine.

For classes, self-study, and the next topic a learner needs.

A budget built around learners

PHP 100,000

60%

for learning review, access and evaluation

Learner access stays free.

Allocation	PHP
Learning review	24,000
Access support	18,000
Evaluation	18,000
Deployment	16,000
Product operations	8,000
Teacher onboarding	8,000
Recruitment content	3,000
Contingency	5,000
Total	**100,000**

backtrack.

A route back to today's lesson.

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Research behind the approach

Khan Academy teacher reports

ALEKS custom objectives

Gemini study notebooks · 2026

IES learning practice guide

Spacing and retrieval · 2022

Confidence assessment · Foster

Self-Determination Theory in education